GeoLocator-AI

Evaluating and Improving VLM Reasoning Reliability in Real-World Geospatial Inference

Executive Summary

This project built a geolocation pipeline that analyzes Street View imagery using Claude Sonnet's vision API to predict geographic coordinates. The system extracts 20 structured features from each image, generates a direct location inference, and stores all results in DuckDB for analytical querying. Across 607 labeled Street View images collected from 17 global regions, the pipeline achieved an average score of 3,490 points per round (out of 5,000 maximum) with a median of 4,052. In two live GeoGuessr games, the system scored 19,379 and 20,740 out of 25,000 across five rounds. While a raw VLM query can produce a one-time location guess, it provides no structured evidence, no stored results, no systematic error analysis, and no mechanism for iterative improvement. This pipeline transforms a single-use inference into a measurable, auditable, and improvable analytical workflow. For instance, the pipeline's structured analysis across 607 rounds revealed that Claude misidentifies South African locations as the American Southwest 62% of the time. This systematic bias would be invisible from individual queries but immediately actionable once surfaced through aggregate analysis.

The project tested two architectural approaches. A Bayesian feature-scoring system averaged 1,025 points with 43% of guesses landing on the wrong continent. Replacing this with direct VLM inference improved performance by 240%. The analytical framework produced several findings: VLM confidence scores are linearly calibrated with actual accuracy, semi-arid Southern Hemisphere landscapes are systematically misidentified as the American Southwest, and prompt instructions placed as post-guess validation checks are more effective than background knowledge sections.

Introduction

GeoGuessr is a browser-based game that places players at random Google Street View locations and challenges them to determine where in the world they are. Each round awards up to 5,000 points based on the distance between the player's guess and the actual location, with five rounds per game totaling a maximum of 25,000. The game serves as a practical benchmark for visual geolocation: an image goes in, coordinates come out, and the scoring formula provides an objective accuracy metric.

The goal of this project was to build a structured pipeline that could consistently score 20,000 or higher per game (an average of 4,000 points per round) while producing auditable, queryable results. The system needed to not just guess correctly, but it also needed to explain why it guessed the way it did and store structured evidence for systematic analysis of error patterns.

Data Collection

The dataset consists of 607 locations collected via the Google Street View Static API. Each location was photographed at four compass headings (0°, 90°, 180°, 270°), producing approximately 2,400 total images. Locations were stratified across 17 geographic regions to prevent overrepresentation of well-covered areas like the United States and Western Europe.

A tiered snap-radius approach addressed the fundamental challenge of sparse Street View coverage in certain regions. Dense-coverage regions (Western Europe, Japan, South Korea) used a 1km snap radius, medium-coverage regions (Brazil, Southeast Asia, India, Eastern Europe) used 10km, and sparse-coverage regions (Russia, sub-Saharan Africa, Middle East, Australia) used 50km. The metadata API's radius parameter snaps to the nearest available panorama, ensuring that collected locations reflect actual Street View coverage patterns rather than geographic centroids.

The resulting dataset is deliberately harder than typical GeoGuessr gameplay. Only 16% of the collected images contain any readable text, compared to an estimated 40–50% in standard GeoGuessr rounds. This rural-heavy distribution means that the pipeline's performance on this dataset represents a lower bound; live game scores with more urban, text-rich images are consistently higher.

Pipeline Architecture

The final pipeline makes a single Claude Sonnet API call per round. This call simultaneously extracts approximately 20 structured geographic features and generates a direct location guess with coordinates, reasoning, and a self-assessed confidence score. The structured features including script type, language, driving side, biome, architecture, road markings, and vegetation are stored alongside the guess for post-hoc analysis but do not influence the location inference.

The 20 features are organized into two passes. Pass 1 covers deterministic signals that can identify a country directly: script type, language, readable text, place names, route numbers, plate format, driving side, speed sign format, domain extensions, and currency symbols. Pass 2 covers probabilistic signals that narrow the region: biome, vegetation species, sky condition, terrain, soil color, architecture style, pole type, road surface, road markings, and infrastructure quality. These features were selected from a reference taxonomy of 127 geographic indicators, filtered to the subset that Claude can reliably extract from Street View imagery without hallucination. Features prone to hallucination such as shadow angles, satellite dish tilt, and camera generation were excluded after testing.

A geocode override path exists for rounds where the extractor identifies a readable place name with high language confidence (above 0.8). In these cases, the pipeline geocodes the place name via Nominatim and uses those coordinates instead of Claude's direct guess. A sanity check ensures the geocoded result is within 100km of Claude's own estimate, preventing mismatches from ambiguous place names. The geocode path fires on approximately 1-2% of rounds but averages 4,734 points when it does.

All results are stored in DuckDB with the full feature set, Claude's reasoning, the raw API response, guess and actual coordinates, and the computed GeoGuessr score. This enables SQL-based analysis of error patterns across the entire dataset.

The pipeline also implements a confidence-based decision system. When Claude's self-assessed confidence exceeds 0.7, the direct inference coordinates are accepted with high trust. When a geocode-able place name is detected with language confidence above 0.8, the geocode path overrides with verified coordinates. For low-confidence rounds (below 0.3), the system's analytical layer flags these for pattern analysis, feeding the iterative prompt refinement loop. This tiered approach ensures that the system adapts its resolution strategy based on inference reliability rather than applying a uniform method to all rounds.

The Bayesian Experiment

The initial architecture decomposed the geolocation task into discrete steps: extract features, score each feature against a Bayesian likelihood table covering 18 world regions, normalize via softmax, and resolve the top-scoring region to a centroid coordinate. A confidence gate assigned each round to a high, medium, or low tier, with low-confidence rounds falling back to biome-aware hedge centroids to prevent overcommitment.

This approach was tested on all 607 images. The results were poor: an average of 1,025 points per round with a median of 350. Forty-three percent of rounds guessed the wrong continent entirely, with 259 rounds exceeding 5,000km of error. Only 66 rounds (11%) landed within 500km of the actual location.

Key finding: Decomposing a VLM's visual understanding into discrete categorical features and reassembling them via likelihood multiplication consistently degrades accuracy compared to the model's direct inference. The Bayesian layer acted as a lossy compression step. Claude correctly identified visual features but the scoring layer failed to weight and combine them as effectively as Claude's own reasoning. This finding motivated the architectural pivot to direct inference.

Architectural Pivot to Direct Inference

The next architecture iteration merged feature extraction and location inference into a single Claude call. Rather than extracting features and feeding them to a separate scoring layer, the prompt asks Claude to analyze the image, extract structured features, and then provide its best location guess with coordinates and reasoning all in one response.

The improvement was immediate and substantial. On the same 607-image dataset, this pipeline averaged 3,490 points with a median of 4,052. A direct comparison on the same image (a rural French road) showed the Bayesian approach scoring 142 points versus the direct inference approach scoring 3,493 points. Fifty-one percent of rounds scored above 4,000 points, and the proportion scoring below 500 dropped from 51% to 9%.

Metric	Bayesian	Direct	Change
Average score	1,025	3,490	+240%
Median score	350	4,052	+1,058%
Rounds > 4,000 pts	47 (8%)	309 (51%)	+558%
Rounds < 500 pts	308 (51%)	53 (9%)	-83%
Wrong continent (>5,000km)	259 (43%)	28 (5%)	-89%

Prompt Refinement

With the architecture validated, the primary remaining optimization lever was the extraction prompt itself. The refinement methodology follows a structured loop: run all images, analyze error patterns in DuckDB, identify systematic misidentifications, add corrective instructions to the prompt, and re-run to measure improvement. This is prompt optimization based on empirical error analysis, not model fine-tuning.

Error-Driven Corrections

Analysis of the direct inference baseline results revealed several systematic failure patterns. The most prominent was the misidentification of South African locations as the American Southwest (62% error rate for South Africa). Claude consistently interpreted dry semi-arid terrain with sparse scrubland as indicative of Texas, Wyoming, or New Mexico, regardless of hemisphere. Corrective instructions were added directing Claude to check for left-hand driving, fence styles, and road marking patterns specific to southern Africa.

Additional corrections addressed Vietnam/Thailand confusion (58% of Vietnamese locations guessed as Thai), Nordic country differentiation (Swedish roads guessed as Finnish 50% of the time), and Mediterranean Europe disambiguation (Italy confused with Portugal and Spain).

The Decisive Rules Discovery

A significant finding emerged from the Mediterranean disambiguation effort. Italy has a nearly unique road marking pattern in Europe: white outer edge lines with no center line. The pipeline correctly extracted this feature ("road_markings: white edge lines, no center line") but Claude's location guess consistently ignored it, defaulting to Portugal based on broader Mediterranean vegetation signals.

Adding the rule "Italy has white edge lines with no center line" to the background knowledge section (Section 8 of the prompt) had no effect. Claude acknowledged the feature in its extraction but did not apply it during inference. However, restructuring the same rule as a post-guess validation check "BEFORE FINALIZING YOUR GUESS, check: did you observe no center line + white edge lines? If yes, change your guess to Italy" caused Claude to correctly override its initial Portugal guess.

Finding: Prompt instructions embedded as background knowledge are less effective than the same instructions framed as post-inference validation checks. The VLM can extract discriminating features correctly but fails to apply them during inference unless forced to explicitly verify its guess against decisive rules. This has implications for prompt architecture in any VLM pipeline: place critical decision rules at the decision point, not in the preamble.

Refinement Impact

The combined prompt refinements (error-driven corrections plus decisive rules) produced a modest improvement on the rural training dataset: average score moved from 3,458 to 3,490 (+32 points). This marginal gain reflects the dataset's composition with 84% rural images lacking text, many remaining failures involve genuinely ambiguous locations where the visual features overlap across continents (e.g., eucalyptus plantations in both Thailand and Brazil). Further prompt refinement faces diminishing returns on this dataset.

Analytical Findings

Feature Importance

The 20 extracted features vary significantly in their impact on scoring accuracy. Biome was the most consistently available feature (98% detection rate) and showed the largest gap between present and absent: 3,511 average points when detected versus 562 when null. Driving side, available on 67% of rounds, produced a 734-point differential (3,665 vs. 2,931). Language detection, while available on only 34% of rounds, correlated with the highest average score when present (3,729).

Plate format was rarely detected (5% of rounds) but highly valuable when found (4,108 average). Place name detection was the most impactful single feature: only 3% of rounds had a geocodable place name, but those averaged 4,490 points. This confirms the pipeline's geocode override path as the highest-accuracy route when available.

Confidence Calibration

Claude's self-assessed confidence score (0.0–1.0) proved to be genuinely predictive of actual accuracy. Rounds with confidence below 0.2 averaged 745 points. Rounds with confidence above 0.9 averaged 4,893 points. The relationship is near-linear across all bins, with no significant miscalibration at any confidence level. This finding is uncommon in VLM evaluations and suggests that Claude's internal uncertainty estimation is reliable for geolocation tasks.

Confidence Bin	Rounds	Avg Score	Median Score
0.0 – 0.2	13	745	75
0.2 – 0.3	43	2,512	3,453
0.3 – 0.4	162	3,164	3,518

0.4 – 0.5	150	3,519	4,134
0.5 – 0.6	118	3,725	4,116
0.6 – 0.7	55	3,906	4,273
0.7 – 0.8	37	4,051	4,558
0.8 – 0.9	18	4,594	4,715
0.9 – 1.0	11	4,893	4,931

Geographic Confusion Patterns

The confusion analysis revealed that cross-continental misidentification is concentrated in a small number of country pairs that share similar visual environments. South Africa was the most misidentified country in the dataset, with 62% of rounds guessed incorrectly primarily as the United States (4 rounds), Botswana (3), and Namibia (2). The root cause is visual similarity between the Karoo semi-desert and the American Southwest: both feature sparse scrubland, paved highways with white markings, and flat terrain extending to distant mesas.

Vietnam and Thailand present a similar challenge: 14 of 24 Vietnamese rounds were guessed as Thailand. Both countries share tropical vegetation, similar road infrastructure, and (from certain angles) indistinguishable landscapes. The key discriminator, Vietnamese Latin script with distinctive diacritics versus Thai's unique script, is only useful when text is visible, which occurs in a minority of rural rounds.

Canada was guessed as the United States on 13 of 18 rounds (72%). This is arguably correct at a practical level since the visual difference between rural Ontario and rural Minnesota is minimal, but it represents a systematic inability to distinguish the two countries without Canadian-specific markers like bilingual signage or Trans-Canada Highway shields.

VLM Geographic Bias

A pattern emerged in the failure analysis: Claude demonstrates a systematic bias toward guessing the United States for arid and semi-arid landscapes regardless of hemisphere or continent. Of the 28 rounds with wrong-continent guesses (>10,000km error), 9 were guessed as the United States. Locations in South Africa, Iran, South Korea, and Tunisia were all attributed to American states (Texas, Wyoming, New Mexico, West Virginia).

This likely reflects the distribution of arid-landscape imagery in Claude's training data, where the American Southwest is heavily represented. The bias persisted even after explicit prompt instructions directing Claude to consider Southern Hemisphere alternatives for dry terrain. This finding has implications for any VLM-based geolocation system: training data distribution creates geographic priors that prompt engineering can mitigate but not eliminate.

Text Detection Impact

In the Bayesian architecture, text detection was critical: rounds with readable text averaged 4,730 points versus 982 without. In V2 direct inference, this gap narrows dramatically: 3,542 with text versus 3,440 without, which is only a 100-point difference. This indicates that direct VLM inference extracts substantial geographic signal from environmental features alone, reducing dependency on text. The practical implication is that the direct inference architecture degrades more gracefully on text-free rural imagery.

Live Validation

The pipeline was validated on actual GeoGuessr gameplay using the Streamlit application interface. Across two live games, the pipeline scored 19,379 and 20,740, with the latter exceeding the project's 20,000-point target. In the first five-round game, the pipeline scored 19,379 out of 25,000, with individual round scores of 4,057, 4,460, 4,842, 1,218, and 4,802. The weak round (1,218 points) involved a Mediterranean road in southern Italy that was misidentified as Portugal, the Italy road-marking confusion pattern that was subsequently addressed through the decisive rules prompt fix. The second game scored 20,740, with individual rounds of 2,583, 4,964, 4,992, 3,210, and 4,991. The weakest round involved interior Brazil, where the pipeline correctly identified the country but placed the guess approximately 1,000km from the actual location which is a common challenge given the visual similarity across Brazil's vast cerrado region.

The same five GeoGuessr screenshots used for initial manual analysis during the planning phase were re-run through the pipeline. The pipeline’s guesses matched or improved upon the manual analysis on three of five rounds, correctly identifying Sisteron (France), Lima (Peru), and Point Reyes Station (California). The two remaining rounds, Chittagong (Bangladesh, guessed as Vietnam) and Huis Ten Bosch (Japan, guessed as Netherlands), produced the same errors as manual analysis, confirming that these are inherent limitations of single-frame geolocation rather than pipeline deficiencies.

Limitations and Future Work

The pipeline has several documented limitations. Replica environments, such as the Huis Ten Bosch Dutch theme park in Japan, present architecturally accurate foreign scenery that is indistinguishable from the real country in a single frame. No single-image system can reliably detect these without external metadata.

Rural images without any text or distinctive infrastructure represent the accuracy ceiling for prompt-based optimization. When a eucalyptus plantation on red laterite soil looks identical in Thailand and Brazil, no amount of prompt refinement will resolve the ambiguity. Multi-point sampling, collecting images from multiple positions along a road to increase the probability of encountering text or distinctive features, is the most promising architectural improvement for these cases.

The 607-image dataset, while sufficient for identifying major patterns, is too small for robust statistical conclusions about individual country-level accuracy. Regions with fewer than 10 samples (Peru, Ethiopia,

Denmark) produce unreliable metrics. A larger dataset focused on underrepresented regions would improve analytical confidence.

Lastly, the pipeline was not tested on countries without Google Street View coverage, including China and North Korea. While Claude can infer these locations from its training knowledge, accuracy for uncovered countries is unmeasured.

Additional improvements include model ensembling (running the same image through multiple VLMs and averaging coordinates), automated GeoGuessr gameplay via browser automation, and expansion of the prompt's geographic knowledge base with region-specific discriminators from competitive GeoGuessr resources.

Real-World Implications

The methodology and findings from this project extend to several operational domains. In open-source intelligence and defense analysis, the ability to determine the geographic origin of an image or video is a routine analytical task. This pipeline demonstrates a structured approach to that task: extract visual evidence, generate a location hypothesis, store the reasoning chain, and measure accuracy against ground truth. The confidence calibration finding that VLM self-assessed confidence is linearly predictive of actual accuracy is directly relevant to any workflow where analysts need to assess the reliability of an automated geolocation estimate.

In fraud detection and content verification, businesses face the challenge of verifying that seller locations, listing photos, and user-submitted content match claimed geographic origins. A structured extraction pipeline can flag inconsistencies for example, a listing claiming to be in Australia but showing right-hand driving and yellow center lines. The feature extraction layer provides the explainable evidence trail that compliance workflows require.

In disaster response and humanitarian operations, rapid geolocation of damage imagery from social media or field reports can accelerate resource allocation. The pipeline's ability to process images without text (84% of the training dataset) is relevant here, as disaster imagery rarely contains readable signage. The finding that direct VLM inference degrades gracefully on text-free imagery, only a 100-point scoring difference, suggests these models can provide useful geographic estimates even from degraded or partial imagery.

Conclusion

This project demonstrates that a structured pipeline built around VLM direct inference can achieve competitive geolocation performance (3,490 average on rural imagery, ~20,000 in live gameplay) while providing structured, query-able, and auditable results that a raw model query cannot.

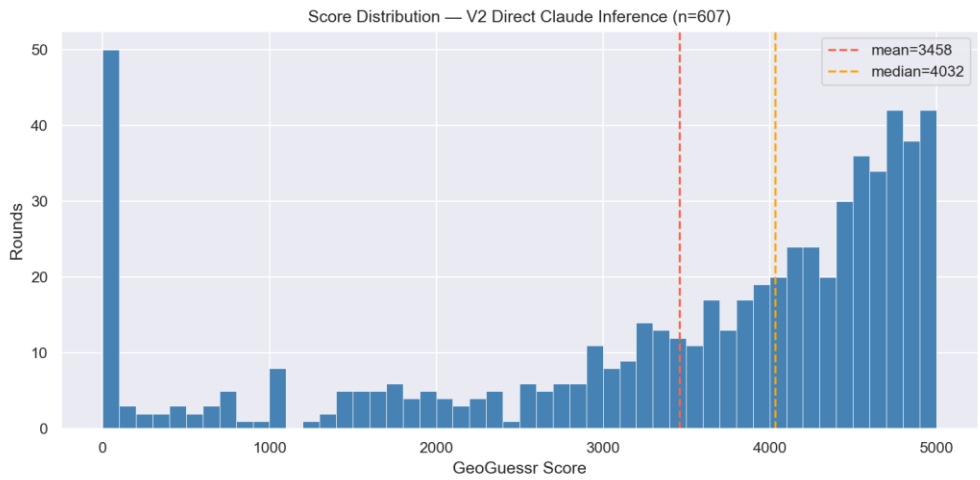
The central technical finding is that VLMs should be guided, not decomposed. Extracting a model's visual understanding into discrete features and reassembling them through external scoring logic consistently

underperforms the model's own inference. The pipeline's value lies not in replacing the model's reasoning but in structuring its output for analysis, storing results for systematic querying, and iteratively refining the prompt based on empirical error patterns.

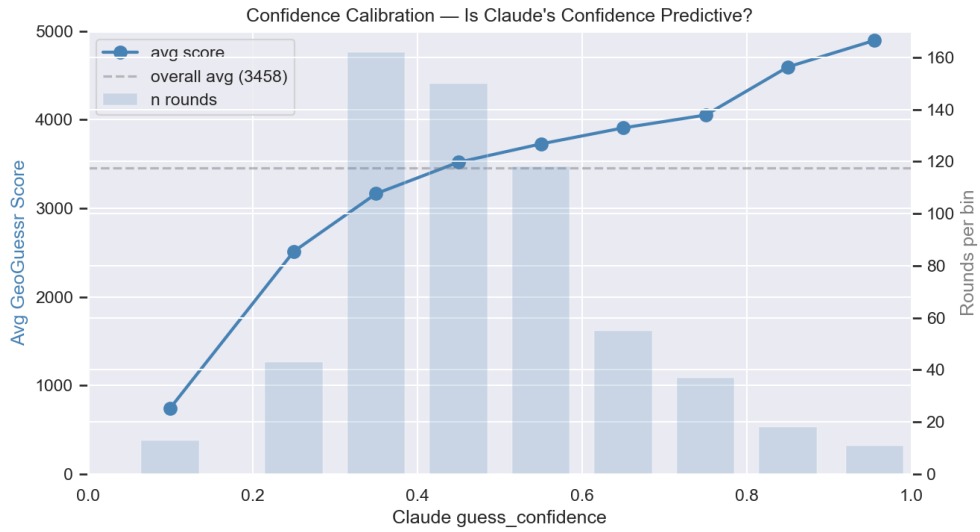
The analytical contributions, confidence calibration curves, geographic confusion patterns, feature importance rankings, and the decisive-rules prompting discovery, provide generalizable insights for any system built on VLM inference. These findings extend beyond geolocation to the broader question of how to effectively operationalize large vision models in structured analytical workflows.

Appendix

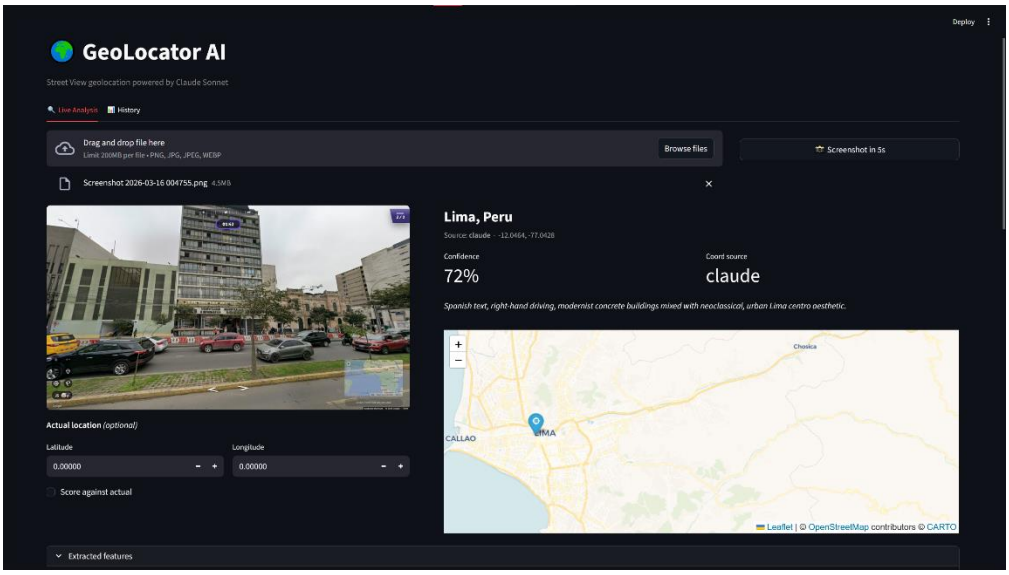
Appendix A: Score distribution chart

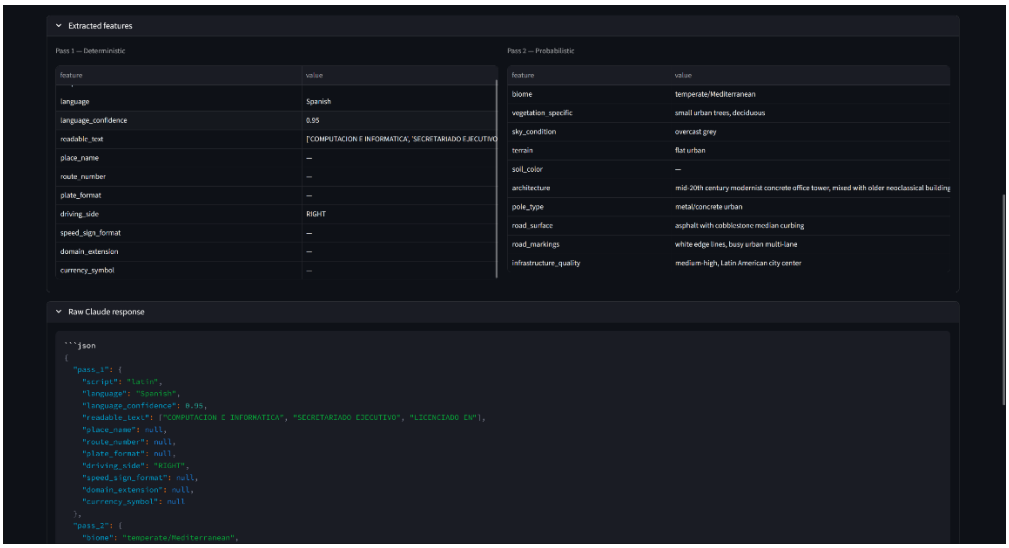


Appendix B: Confidence calibration chart

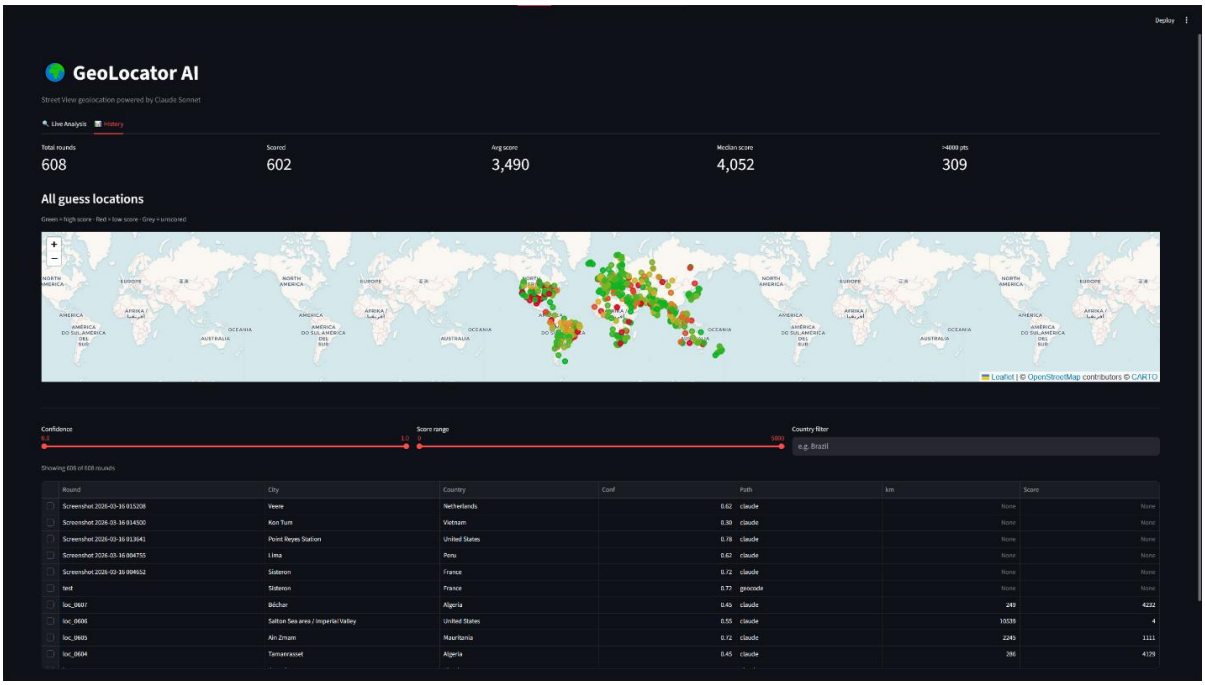


Appendix C: App live analysis view

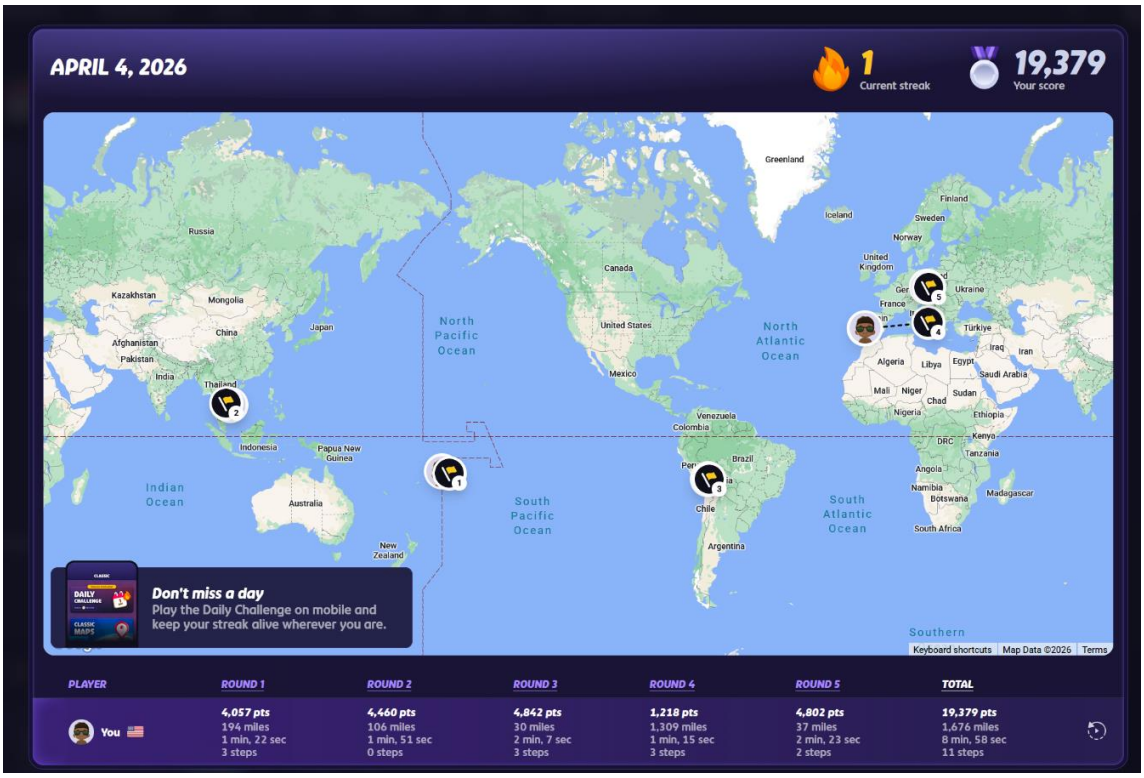




Appendix D: App history view



Appendix E: Live validation results



APRIL 6, 2026
1
Current streak
20,740
Your score
Greenland
Iceland
Finland
Sweden
Norway
Russia
Canada
United Kingdom
Ireland
Germany
France
Spain
Italy
Ukraine
Kazakhstan
Mongolia
United States
Türkiye
China
Japan
Iraq
Iran
Afghanistan
Pakistan
Algeria
Libya
Egypt
Saudi Arabia
India
Mexico
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Venezuela
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Indonesia
Papua New Guinea
Peru
Brazil
Bolivia
Angola
Namibia
Botswana
Madagascar
Australia
Chile
South Africa
Argentina
North Atlantic Ocean
North Pacific Ocean
South Pacific Ocean
South Atlantic Ocean
Indian Ocean
Southern Ocean
Don't miss a day
Play the Daily Challenge on mobile and keep your streak alive wherever you are.
Keyboard shortcuts
Map Data ©2026
Terms
PLAYER
ROUND 1
ROUND 2
ROUND 3
ROUND 4
ROUND 5
TOTAL
You
2,583 pts
612 miles
1 min, 42 sec
0 steps
4,964 pts
6.7 miles
2 min, 23 sec
1 step
4,992 pts
1.4 miles
1 min, 35 sec
4 steps
3,210 pts
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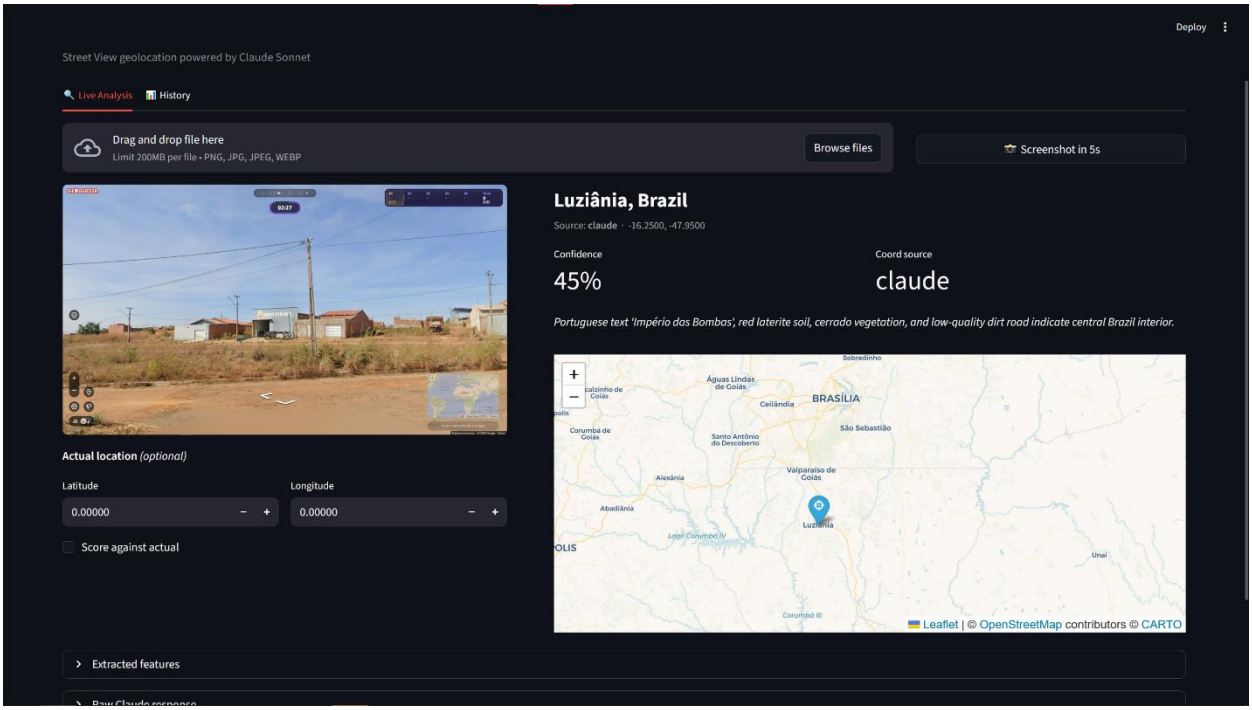
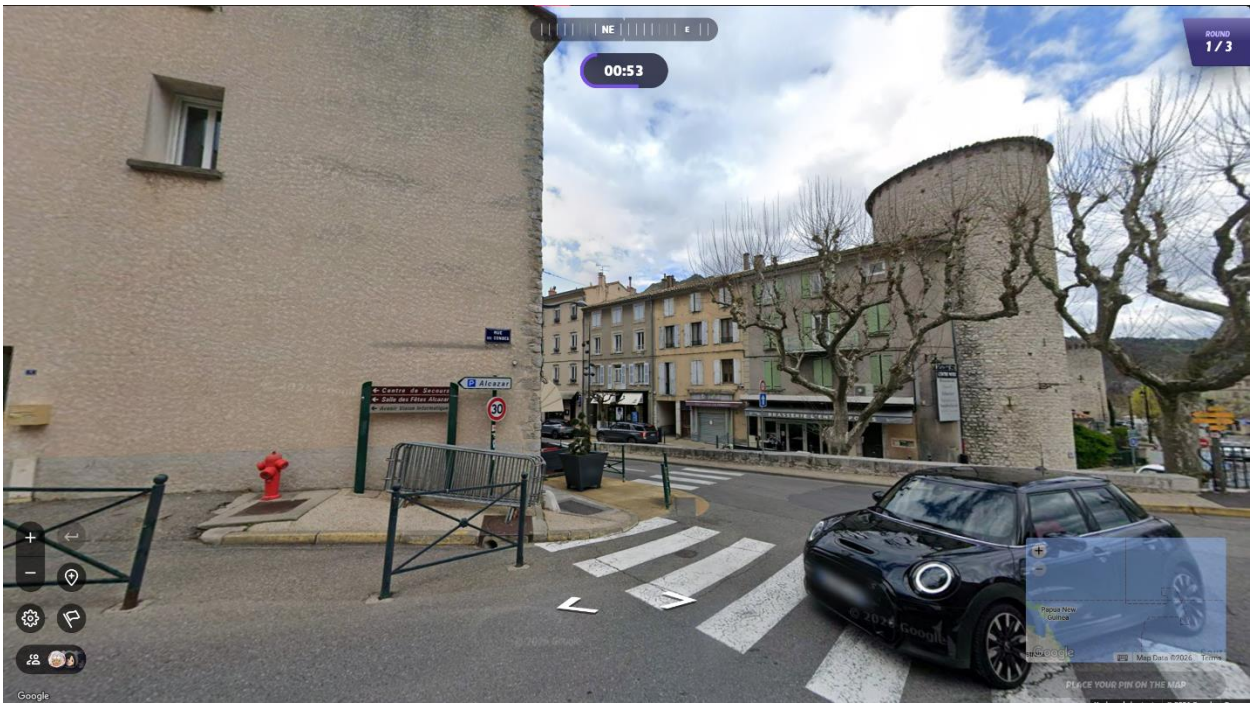
9 min, 23 sec
6 steps


Deploy
Street View geolocation powered by Claude Sonnet
Live Analysis
History
Drag and drop file here
Limit 200MB per file • PNG, JPG, JPEG, WEBP
Browse files
Screenshot in 5s
Luziânia, Brazil
Source: claude · -16.2500, -47.9500
Confidence
45%
Coord source
claude
Portuguese text 'Império das Bombas', red laterite soil, cerrado vegetation, and low-quality dirt road indicate central Brazil interior.
Actual location (optional)
Latitude
0.00000
Longitude
0.00000
Score against actual
Sobradinho
Águas Lindas de Goiás
Cidalvinho de Goiás
Ceilândia
BRASÍLIA
Corumbá de Goiás
São Sebastião
Santo Antônio do Descoberto
Valparaíso de Goiás
Alexânia
Abadiânia
Luziânia
Unaí
Leaflet | © OpenStreetMap contributors © CARTO
Extracted features

Appendix F: Manual test images



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